

# Anto Idicherian Lonappan

✉ antolonappan@icloud.com

☎ +1 (858) 319-5166



## Research Interests

- ◇ Gravitational lensing of the Cosmic Microwave Background(CMB)
- ◇ CMB data analysis
- ◇ Application of Machine Learning in Cosmology
- ◇ Observational Cosmology
- ◇ Data Science

## Affiliations

- ◇ Department of Astronomy & Astrophysics, School of Physics, University of California, San Diego.
- ◇ IFPU - Institute for Fundamental Physics of the Universe

## Education

- 2018 – 2022 ◇ **Ph.D in Astrophysics and Cosmology, Cum Laude(with honors).**  
Scuola Internazionale Superiore di Studi Avanzati-SISSA  
Thesis title: *Cosmic Microwave Background: Analysis and Interpretation of B Modes for the detection of Inflationary Gravitational Waves and Gravitational Lensing.*
- 2014 – 2016 ◇ **M.Sc. Physics**  
St. Berchmans College, Mahatma Gandhi University  
Thesis title: *Markov Chain Monte Carlo Methods: Cosmological Parameter Estimation*
- 2010 – 2013 ◇ **B.Sc. Physics**  
St. Berchmans College, Mahatma Gandhi University

## Research Experience

- 2024 – Present ◇ **Postdoctoral Research Fellow**, University of California, San Diego.
- 2022 – 2024 ◇ **Postdoctoral Research Fellow**, University of Rome, Tor Vergata.
- 2022 ◇ **Visiting Research Fellow**, Kavli-IPMU University of Tokyo, Japan.
- 2017 – 2018 ◇ **Junior Research Fellow**, Presidency University, India.
- 2016 – 2017 ◇ **Research student**, Centre for Theoretical Physics, JMI, India.

## Certificates

- ◇ Deep Learning and Reinforcement Learning
- ◇ Unsupervised Machine Learning
- ◇ Supervised Machine Learning: Classification
- ◇ Supervised Machine Learning: Regression
- ◇ Introduction to Computer Science and Programming Using Python

## Awards & Grants

---

- ◇ Junior Research Fellowship(2017)-Department of Science and Technology(DST), Government of India.

## Scientific Collaborations

---

- ◇ POLARBEAR
- ◇ Simons Array
- ◇ Simons Observatory
- ◇ LiteBIRD

## Projects

---

- Ongoing
- ◇ **CMB-S4 x LiteBIRD: Multitracer Delensing**  
 <https://github.com/litebird/LBxS4>
  - ◇ **Estimating  $\tau$  from PLANCK using Neural Network**  
 [https://github.com/antolonappan/taunet\\*](https://github.com/antolonappan/taunet*)
  - ◇ **Non linearity in Bolometers: Impact on CMB Bispectra, Lensing and Delensing**  
 [https://github.com/antolonappan/legend\\*](https://github.com/antolonappan/legend*)
  - ◇ **Reducing Dispersion in  $Y_{Ksz} - M$  relation using Machine Learning**  
 [https://github.com/antolonappan/KSZscaler\\*](https://github.com/antolonappan/KSZscaler*)
  - ◇ **Multi-Task Gaussian Process for Cosmological application**  
 <https://github.com/antolonappan/marcia>
- Completed
- ◇ **Delensing LiteBIRD**  
 <https://github.com/antolonappan/dell>  
 <https://github.com/litebird/LiteBIRD-lensing>
  - ◇ **Data Quality analysis of Simon Array**  
 [https://bitbucket.org/berkeleylab/simons\\_array\\_offline\\_software](https://bitbucket.org/berkeleylab/simons_array_offline_software)
  - ◇ **Constrain on  $r$  using POLARBEAR Data**  
 [https://github.com/antolonappan/PB\\_LP\\_final](https://github.com/antolonappan/PB_LP_final)  
 [https://github.com/PolarbearEU/LargePatch\\_reanalysis](https://github.com/PolarbearEU/LargePatch_reanalysis)
  - ◇ **Global effect of AGN Feedback**  
 <https://github.com/antolonappan/ABE-I>
  - ◇ **Bayesian Evidence for Dark Energy models**  
 <https://github.com/antolonappan/cosmopass>

---

\*currently private repository.

## Skills

- Coding     ♦ PYTHON, C, C++, R, MPI, Bash, Zsh,  $\LaTeX$
- Databases   ♦ MySQL, SQLite.
- Web Dev   ♦ HTML, CSS, JavaScript, Apache, Nginx.
- OS       ♦ LINUX and MacOS.

## Referee

- ♦ Astronomy & Astrophysics (A&A)

## Publications

### Major Contributions

- 1 Lonappan, A. I. et al. (2023). LiteBIRD Science Goals and Forecasts: A full-sky measurement of gravitational lensing of the CMB. arXiv: 2312.05184 [astro-ph.CO]
- 2 Namikawa, T., Lonappan, A. I. et al. (2023). LiteBIRD Science Goals and Forecasts: Improving Sensitivity to Inflationary Gravitational Waves with Multitracer Delensing. arXiv: 2312.05194 [astro-ph.CO]
- 3 Adachi, S., Adkins, T., Faú ndez, M. A. O. A., Arnold, K. S., Baccigalupi, C., Barron, D., ... Zhou, Y. (2022). Improved upper limit on degree-scale CMB b-mode polarization power from the 670 square-degree POLARBEAR survey. *The Astrophysical Journal*, 931(2), 101. [doi:10.3847/1538-4357/ac6809](https://doi.org/10.3847/1538-4357/ac6809)
- 4 Kar Chowdhury, R., Chatterjee, S., Lonappan, A. I., Khandai, N., & Di Matteo, T. (2020). Cosmological simulation of galaxy groups and clusters. i. global effect of feedback from active galactic nuclei. *The Astrophysical Journal*, 889(1), 60. [doi:10.3847/1538-4357/ab5b96](https://doi.org/10.3847/1538-4357/ab5b96)
- 5 Lonappan, A. I., Kumar, S., Ruchika, Dinda, B. R., & Sen, A. A. (2018). Bayesian evidences for dark energy models in light of current observational data. *Physical Review D*, 97(4). [doi:10.1103/physrevd.97.043524](https://doi.org/10.1103/physrevd.97.043524)
- 6 Lonappan, A. I., Ruchika, & Sen, A. A. (2017). Is it time to go beyond  $\Lambda$ CDM universe? [doi:10.48550/arXiv.1705.07336](https://doi.org/10.48550/arXiv.1705.07336). arXiv: 1705.07336 [astro-ph.CO]

### In Preparation

- 1 Lonappan, A. I., Pagano, L., & Krachmalnichof, N. (2024). *Estimation of the optical depth to reionization  $\tau$  using CNN*.
- 2 Lonappan, A. I., Ghinga, T., Namikawa, T., & Giuseppe, P. (2024). *Detector non-linearities: Impact in CMB science*.
- 3 Lonappan, A. I., & Singh, P. (2024). *Improving the kinetic sunyaev-zel'dovich scaling relation using machine learning*.

### PhD thesis

- 1 Lonappan, A. I. (2022). *Cosmic Microwave Background: Analysis and interpretation of B-modes for the detection of Inflationary Gravitational Waves and Gravitational Lensing*. Retrieved from <https://hdl.handle.net/20.500.11767/129630>

### Other Contributions

- 1 Campeti, P. et al. (2023). *LiteBIRD Science Goals and Forecasts. A Case Study of the Origin of Primordial Gravitational Waves using Large-Scale CMB Polarization*. arXiv: 2312.00717 [astro-ph.CO]

2

Fuskeland, U. et al. (2023). *Tensor-to-scalar ratio forecasts for extended LiteBIRD frequency configurations*.  
[doi:10.1051/0004-6361/202346155](https://doi.org/10.1051/0004-6361/202346155). arXiv: 2302.05228 [astro-ph.CO]